

Md Rysul Kabir

Bloomington, Indiana, USA

[LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Summary

Computer Science Ph.D. student working on LLM safety, reinforcement learning, and Bayesian modeling. Builds and evaluates machine learning systems in Python and PyTorch, with experience in LLM post-training, distributed multi-GPU training, and policy optimization.

Education

Indiana University Bloomington | Bloomington, IN

Ph.D. in Computer Science (in progress); GPA: 3.76/4.0 | Aug 2021 – Present

University of Texas Rio Grande Valley | Edinburg, TX

M.S. in Computer Science; GPA: 4.0/4.0 | Aug 2019 – Jul 2021

Bangladesh University of Engineering and Technology | Dhaka, Bangladesh

B.S. in Mechanical Engineering; GPA: 3.71/4.0; Rank: 17/202 | Feb 2013 – Sep 2017

Technical Skills

Languages: Python, C++, C, SQL, R, MATLAB.

Methods: Deep learning, reinforcement learning, natural language processing (NLP), hierarchical Bayesian modeling, LLM safety evaluation.

LLM Post-Training: Supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), reinforcement learning with verifiable rewards (RLVR).

Frameworks and Libraries: PyTorch, Hugging Face Transformers, NumPyro, JAX, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn.

Systems and Tools: VERL, FSDP, vLLM, Git, Linux (Bash/Zsh), Slurm/HPC, Docker, GCP Compute Engine, MySQL, PostgreSQL, LaTeX.

Research Experience

Indiana University Bloomington | Bloomington, IN

Research Assistant (AI, Cognitive Computing, and LLMs) | Aug 2021 – Present

- Compared harmful SFT, RLVR, and refusal-feature ablation across open-weight LLMs using behavioral, capability, and self-audit evaluations; identified distinct failure modes despite similarly high harmful compliance. [Paper](#)
- Developed a deep RL agent with time-scale invariant memory for temporal decision-making; published at **AAAI 2025**. [Project](#) | [Code](#)
- Built hierarchical Bayesian models in NumPyro to infer path-integration error parameters and compare subjective cognitive decline (SCD) and control groups; published in **Science Advances**. [Code](#)

University of Texas Rio Grande Valley | Edinburg, TX

Research Assistant (Reinforcement Learning) | Aug 2019 – Jul 2021

- Proposed Saltatory Rewards and integrated generalized advantage estimation (GAE) into A2C, achieving **1.5x faster convergence** in reference-based humanlike motion learning. [Thesis](#)
- Implemented PPO, TRPO, A2C, and DQN from scratch in TensorFlow and benchmarked them against standard baselines. [Policy-gradient code](#) | [TD-learning code](#)

Publications

Peer-Reviewed Journal and Conference Papers

1. B. Dickson, J. Mochizuki-Freeman, **M. R. Kabir**, and Z. Tiganj. "Time-Local Transformer." *Computational Brain & Behavior*, Special Issue: Sensory Prediction: Engineered and Evolved, 2026. [Paper](#)
2. **M. R. Kabir**, J. Mochizuki-Freeman, and Z. Tiganj. "Deep Reinforcement Learning with Time-Scale Invariant Memory." *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI)*, 2025. [Paper](#)
3. V. Segen, **M. R. Kabir**, et al. "Path Integration Impairments Reveal Early Cognitive Changes in Subjective Cognitive Decline." *Science Advances*, 2025. [Paper](#)
4. J. Mochizuki-Freeman, **M. R. Kabir**, and Z. Tiganj. "Incorporating a Cognitive Model for Evidence Accumulation into Deep Reinforcement Learning Agents." *Proceedings of the Annual Meeting of the Cognitive Science Society*, 2024. [Paper](#)

Workshop Paper

J. Mochizuki-Freeman, **M. R. Kabir**, M. Gulecha, and Z. Tiganj. "Geometry of Abstract Learned Knowledge in Deep RL Agents." *NeurIPS Workshop on Symmetry and Geometry in Neural Representations*, 2023. [Paper](#)

Preprint

M. R. Kabir and Z. Tiganj. "Different Paths to Harmful Compliance: Behavioral Side Effects and Mechanistic Divergence Across LLM Jailbreaks." *arXiv preprint arXiv:2604.18510*, 2026. [Paper](#)

Master's Thesis

M. R. Kabir. "Effects of Saltatory Rewards and Generalized Advantage Estimation on Reference-Based Deep Reinforcement Learning of Humanlike Motions." *Master's thesis, University of Texas Rio Grande Valley*, Theses and Dissertations, 899, 2021. [Thesis](#)

Poster Presentations

- "Deep Reinforcement Learning with Time-Scale Invariant Memory." *39th AAAI Conference on Artificial Intelligence*, 2025.
- "Towards Time-Scale Invariant Reinforcement Learning." *Cognitive Computational Neuroscience (CCN)*, 2024.
- "Identifying Sources of Path Integration Error Using Bayesian Hierarchical Modeling." *Interdisciplinary Navigation Symposium (iNAV)*, 2024.
- "Making Deep Neural Networks Scale-Invariant Using Cognitive Models." *Context and Episodic Memory Symposium (CEMS)*, 2024.

Selected Technical Projects

LLM Post-Training with VERL | *Python, PyTorch, FSDP, vLLM, Slurm*

- Extended VERL with custom rollout and reward logic for multi-GPU SFT/RLVR, using Fully Sharded Data Parallel (FSDP) and vLLM.

COVID-19 Literature Mining and Clustering | *Python, scikit-learn, NLP*

- Analyzed more than 340,000 COVID-19 articles using TF-IDF, PCA/t-SNE, K-Means/GMM, and LDA/NMF to identify recurring research themes.

Gaussian Process Regression and Kernel Methods | *Python, NumPy*

- Implemented Gaussian process regression from scratch with RBF and polynomial kernels; optimized hyperparameters through log marginal likelihood maximization.

Adversarial Game-Playing Agent (Raichu) | *Python, search algorithms*

- Built a game-playing agent using Minimax, alpha-beta pruning, and custom board-evaluation heuristics to make decisions under strict time limits.

Humanoid Biped Gym Environment | *Continuous control, simulation*

- Built a continuous-control environment with 12-dimensional Box action and 32-dimensional Box observation spaces. [Code](#)

Home Needs Services Marketplace | *Full-stack development*

- Developed a service marketplace with separate frontend and backend applications. [Frontend](#) | [Backend](#)

Teaching Experience

Indiana University Bloomington | Bloomington, IN

Associate Instructor | Aug 2021 – May 2026

- Served as head teaching assistant for Data Mining; authored homework and exam questions and coordinated grading.
- Supported Machine Learning, Applied Machine Learning, and Applied Algorithms through office hours, grading, and student guidance.

University of Texas Rio Grande Valley | Edinburg, TX

Teaching Assistant | Aug 2020 – Jul 2021

- Served as lead instructor for Introduction to Computer Science and as a grader for Object-Oriented Programming in Java.

Academic Service

Peer Reviewer, Neural Information Processing Systems (NeurIPS) | 2026

Peer Reviewer, Cognitive Computational Neuroscience (CCN) | 2024

Selected Awards and Honors

Presidential Graduate Research Assistantship Award. University of Texas Rio Grande Valley | 2019 – 2021

First Runner-Up, MechBlitz Project Show. Bangladesh University of Eng. and Tech. | 2015

Dean's List, Department of Mechanical Engineering. Bangladesh University of Eng. and Tech. | 2013